10.5281/zenodo.21311833

Biologized Grid-space World Model (BGWM): Bottom-Up Cognitive Cascade and Allocentric Spatial Representation in Robotic Control

Concept Draft

Matyas Borda

July 2026
Budapest, Hungary
bordamatyas82@gmail.com

Abstract

The dominant paradigm of current Vision-Language-Action (VLA) robotic systems relies on monolithic, language-based architectures. This approach faces fundamental limitations: human language is a lossily compressed representation that flattens the spatiotemporal dimensions of physical world states, while the egocentric camera view might make planning unstable. In this work, we introduce a biologically inspired, modular world model architecture (BGWM - Biologized Grid-Space World Model) that aims to solve the problems of catastrophic forgetting and spatial hallucinations. Our system is built on three main pillars: (1) the structured decomposition of instructions and vision into Entity Aggregates, (2) an Allocentric Grid-Cell mechanism that creates a topological cognitive map independent of the robot's movement, and (3) a Cognitive Cascade World Model, a bottom-up architecture operating with skip-connections and flexible bottlenecks. The system is trained based on a strictly staged curriculum following biological ontogeny, explicitly separating motor trajectory generation from linguistic reflection.

Keywords: Cognitive Robotics, World Models, Allocentric Representation, Hierarchical Transformers, Object-Centric Learning, Embodied AI

1. Introduction

The most significant challenge in modern robotic control is the integration of physical intelligence and semantic understanding. Current VLA (Vision-Language-Action) models—such as RT-X or Octo—typically pool linguistic, visual, and motor data into a single gigantic Transformer network (LLM/VLM backbone). This paradigm raises two severe issues: *LLM*

Compression Loss: Natural language condenses actions into discrete tokens that lose spatiotemporal information. A robot, however, does not need words, but precise, continuous 3D physics for execution. *Egocentric Myopia:* Models learn from camera images, meaning that if the robot turns, the entire reality "shifts" for the network, which the system must relearn from the entire data stream; otherwise, it leads to a lack of object permanence and spatial understanding. The BGWM (Biologized Grid-Space World Model) breaks away from this monolithic paradigm, drawing inspiration from autonomous machine intelligence concepts [10]. Our solution is the engineering mapping of neurological pathways (dorsal/ventral streams, hippocampus), which aims to ensure the simulation of the physical world without catastrophic forgetting through Three Pillars (Entity Decomposition, Allocentric Grid, Hierarchical Cascading).

2. Related Work

The architecture of BGWM combines several modern research areas. In the field of visual grounding and object-centric learning, models like DINO.txt [1] and Slot Attention [14] provide the foundation for extracting individual entities from raw data. Regarding spatial representation, our work builds upon the classical Mishkin-Ungerleider dorsal-ventral pathway theory [13] and its modern interpretations involving structural cross-talk [4]. The transformation of these egocentric inputs into allocentric spatial grids is heavily inspired by the computational modeling of biological Place Cells and Grid Cells (Banino et al. [9]), mimicking the parieto-medial temporal pathway (PPC to Hippocampus) observed in the brain [5]. Finally, the modular training builds on Hierarchical Vision Transformers [15] and structures that utilize narrowed information channels, contrasting with the currently popular but fragile end-to-end VLM architecture aproach.

3. System Architecture (The Three Pillars)

3.1. Input Stream: Linguistic Atomization and Visual Grounding

Traditional monolithic VLA systems project the raw instruction and image pixels into a single common latent space. The first step of BGWM is the cognitive "decompression" of linguistic compression.

Semantic Atomization: A free-text instruction (e.g., "The bedroom needs to be tidied up because someone threw the socks around…") is received by a dedicated, quantized LLM (e.g., Gemma4:e4B [2]) and processed in three steps: (1) Action extraction, (2) Elementary decomposition into a temporal sequence (Task Queue), and (3) Strict reduction to a Subject-Verb-Object (S-V-O) triplet format (e.g., [Robot Arm] -> [Grasps] -> [Coffee Cup]).

Closed-Loop Visual Grounding: The S-V-O format serves as the bridge between language and vision. The Subject and Object tokens act as prompts for the visual module (utilizing the zero-shot capabilities of DINO.txt [1]), which generates dense heatmap masks from the RGB-D data stream, identifying the target object and the end-effector.

3.2. First Pillar: Entity Aggregates (The Deconstruction of Reality)

To simulate the distinct yet interacting dorsal ("where") and ventral ("what") pathways [4], [13], we must prevent the data from being blended into an undifferentiable vector (conceptually aligning with object-centric learning [14]). We segregate the raw material into three rigid containers: 1. Object Aggregate: Egocentric 3D spatial coordinates and visual features. 2. Body/Arm Aggregate: The spatial pose of the joint, supplemented with the gripper's formation or affordance [12]. 3. Background Aggregate: The static "point cloud" of the environment. *Structured Linking (Entity Tokenization):* These building blocks are packed into a fixed-structure data container with unique [ID] and [TYPE] tokens. An MLP network fuses these into a logically separated representation token.

$$\text{Entity\_Vector} = [[\text{ID}], [\text{TYPE}], X_{ego}, F_{ventral}, A_{affordance}]$$

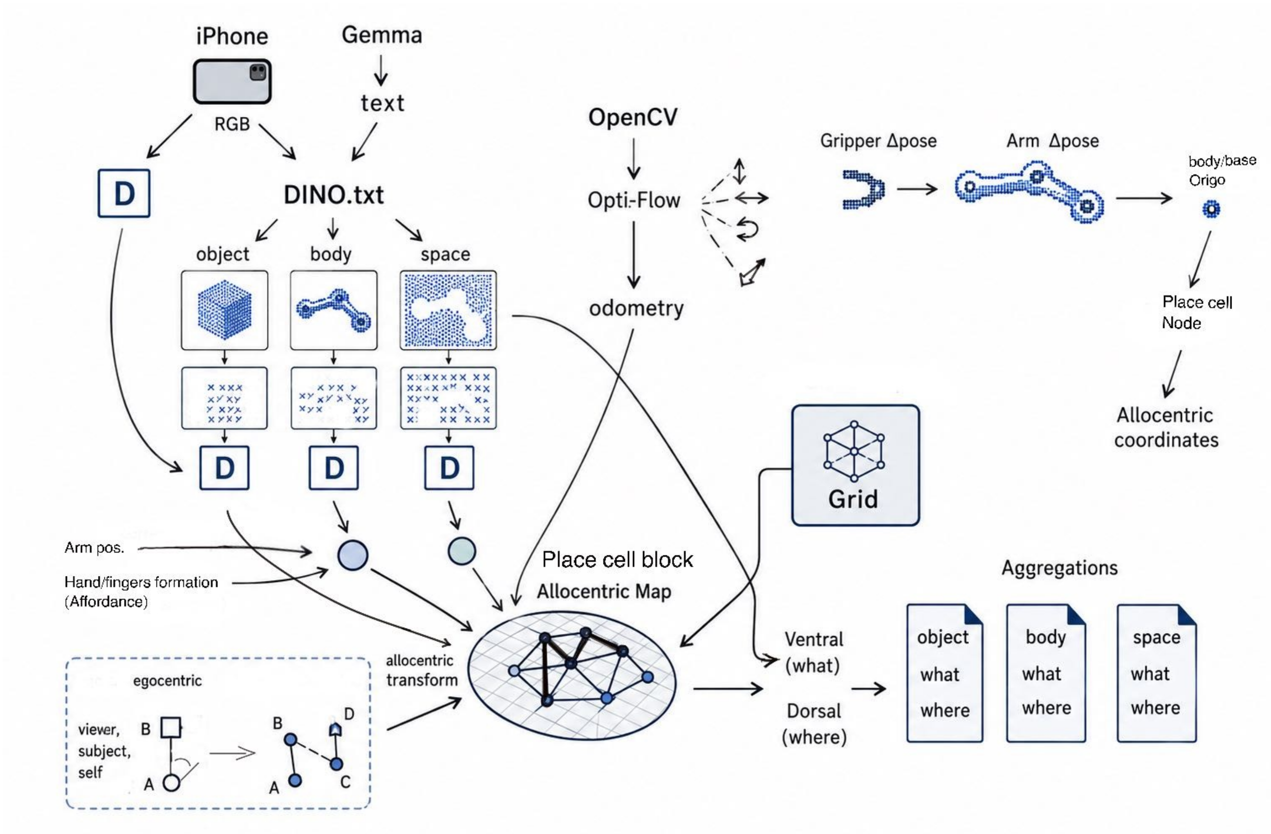

Figure 1: Overview of the Input Stream and Allocentric Transformation. The system processes visual, semantic, and kinesthetic data (Egocentric Inputs), fusing them through the Grid-Cell Binder to create structured Allocentric Entity Aggregates.

3.3. Second Pillar: Allocentric Transformation and the "Place Cell" Spatial Grid

The Grid-Cell Binder module is responsible for rotating egocentric data into absolute space.

Dynamic Coordinate Transformation: Using the inverse of the homogeneous transformation matrix calculated from odometry (derived via standard libraries such as OpenCV [3]), denoted as $T_{odometry}$, we obtain the absolute 3D position of the entities: $X_{allo} = T_{odometry}^{-1} \cdot X_{ego}$.

Topological Binding (Cross-Attention Binding): The nodes of a global virtual grid (G) form the Query. The absolute coordinates (X_{allo}) serve as Keys, and the features serve as Values. The attention mechanism physically "writes" the entity features into the given fixed grid node, creating the pure Allocentric World State.

3.4. Third Pillar: The Cognitive Cascade World Model

Monolithic Transformer backbones can often lead to catastrophic forgetting, especially in a staged training strategy. BGWM instead applies a Bottom-Up Hierarchical Architecture, inspired by hierarchical network designs [15]. The system is built from small, specialized Cognitive Primitive blocks. Through skip-connections, every hierarchical level of the system has access to the raw or semi-processed output of all the levels below it. A Flexible Bottleneck (Perceiver Cross-Attention) at the module's input filters the data from previous levels. Thus, higher levels can select what they need, while the time dimension is threaded together by an internal heartbeat ("Timestamp tokens").

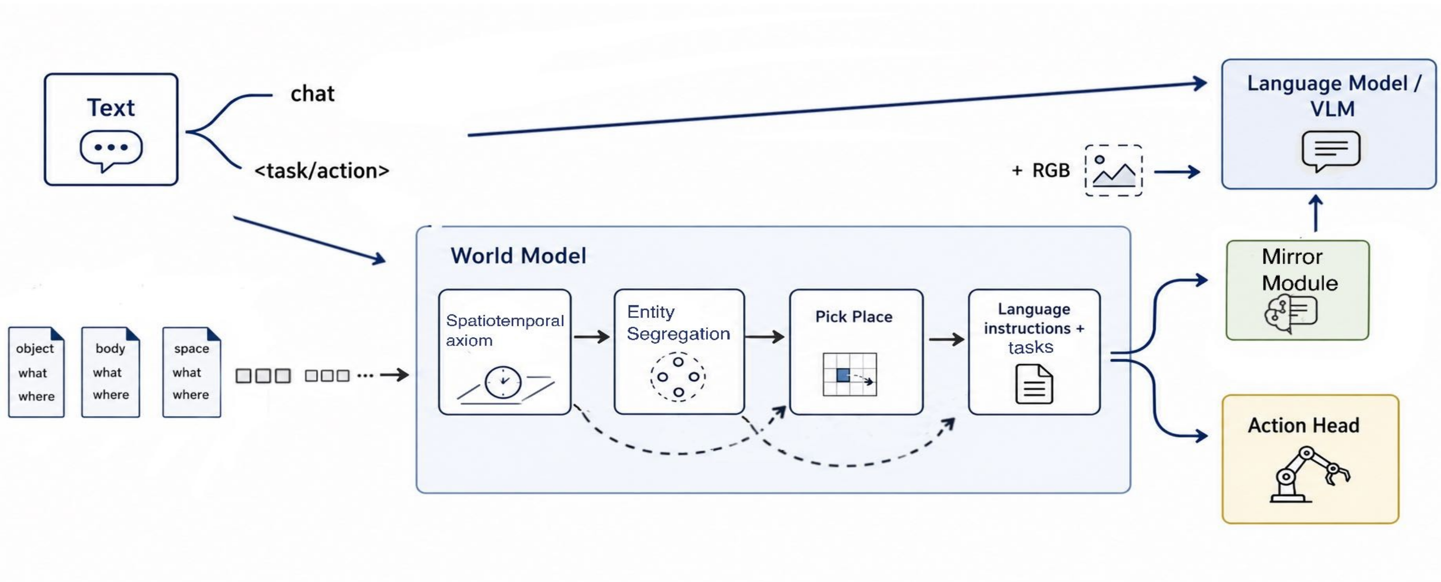

Figure 2: Architecture of the Cognitive Cascade World Model. The bottom-up hierarchical structure utilizes flexible bottlenecks and skip-connections to process spatiotemporal data, ultimately feeding into the dedicated Action and Language heads.

3.5. Output Heads: Action and Reflection

Motor Head (Flow-Matching): Generates a smoothed motion sequence (Action Chunk) from the predictions utilizing Flow Network Matching algorithms [11]. The head is embodiment-agnostic: it predicts the 3D delta positions of the end-effector on the grid rather than joint angles, which can be executed by any IK controller.

Semantic Mirror Module: Intended to reverse the function of the world model to provide linguistic output for the language module. Physical predictions are extremely dense (e.g., 30 Hz). A Mirror Module simulating a biological feedback loop detects events from the world model's output and compresses them into a short, textual factual description (e.g., "Object/Mug grasped").

Language Head (Modular VLM): This condensed textual acknowledgment, along with the visual context and chat history, is fed to a dedicated VLM. The model learns to separate action from chatting, but when necessary, provides an answer rooted in physical reality.

4. The Hierarchical Training Curriculum (Cognitive Ontogeny)

The system employs a strictly staged strategy following biological ontogeny and core knowledge theories [7], without logical loss.

4.1. The Stages of Cognitive Development

1. *Spatiotemporal Axiom:* Integration of odometry and allocentric transformation, space, and displacement.

2. *Entity Segregation:* Separation of static (Background) and dynamic (Object) elements.

3. *Proprioceptive Self-Awareness:* Recognition of the own body (Arm/Gripper) in the context of motor intent, mimicking early sensori-motor development observed in infants [8].

4. *Affordance and Interaction:* Aligning the size/shape of the body and the object (e.g., grasping) based on ecological action potentials [12].

5. *Elementary Intuitive Physics:* Learning gravity, collisions, and object-object interactions.

6. *Causal Sequencing:* Long-term cause-and-effect (e.g., tidying up).

7. *Linguistic Instruction Fine-Tuning:* Connecting language with the already complex physical worldview.

4.2. Addressing the Challenge of Catastrophic Forgetting

When the system learns the N-th logical level, the weight matrices of all previous levels (from 1 to $N-1$) are strictly frozen. The new block uses the Flexible Bottleneck to select from the frozen, stable knowledge. In this way, the entire physical reality is condensed into the highest hierarchical level, yet it can decode and reach back to the most ancient spatiotemporal axioms at any time.

5. Conclusion and Future Work

BGWM concept aims to demonstrate that robotic cognition does not have to lose spatiotemporal dimensions if an appropriate, biologically inspired structured architecture is applied. Future research will focus on two critical areas:

5.1. Computational Efficiency and Predictive Coding:

Current large world models recalculate the entire world state at every time step. In biology, based on the free-energy principle [6], top-down prediction is subtracted from bottom-up sensory data, so the brain only processes the prediction error. Our future goal is to have only these small, delta-changes flow up to the higher layers of the Cognitive Cascade.

5.2. Long-Term Spatial Memory (Episodic Spatial Memory):

Due to the limitations of the context window, the network “forgets” rooms it left minutes earlier. In future work, the contents of the Grid Nodes will be written out to an external, vector database-based Long-Term Memory (RAG). An Internal Attention mechanism can automatically retrieve and load the previously built spatial “grid” into the current context window, creating a true, persistent episodic memory.

References

[1] N. Nguyen et al., “DINO.txt: A Zero-Shot Framework for Text-Guided Visual Grounding,” *arXiv preprint arXiv:2412.16334*, 2024.

[2] Gemma Team et al., “Gemma: Open Models Based on Gemini Research and Technology,” *Google DeepMind*, 2024. [Online]. Available: arxiv.org/abs/2403.08295

[3] G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, vol. 25, pp. 120-123, 2000.

[4] D. J. Kravitz, K. S. Saleem, C. I. Baker, and M. Mishkin, “A new neural framework for visuospatial processing,” *Nature Reviews Neuroscience*, vol. 12, no. 4, pp. 217-230, 2011.

[5] A. Bicanski and N. Burgess, “A neural-level model of spatial memory and imagery,” *eLife*, vol. 7, e33752, 2018.

[6] K. Friston, “The free-energy principle: a rough guide to the brain?,” *Trends in Cognitive Sciences*, vol. 13, no.

7, pp. 293-301, 2009. [7] E. S. Spelke and K. D. Kinzler, “Core knowledge,” *Developmental Science*, vol. 10, no. 1, pp. 89-96, 2007.

[8] C. von Hofsten, “An action perspective on motor development,” *Trends in Cognitive Sciences*, vol. 8, no. 8, pp. 266-272, 2004.

[9] A. Banino et al., “Vector-based navigation using grid-like representations in artificial agents,” *Nature*, vol. 557, no. 7705, pp. 429-433, 2018.

[10] Y. LeCun, “A path towards autonomous machine intelligence,” *OpenReview*, vol. 62, no. 1, 2022.

[11] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow Network Matching: Exact Score Matching with Exact Optimal Transport,” *International Conference on Learning Representations (ICLR)*, 2023.

[12] J. J. Gibson, *The Ecological Approach to Visual Perception*. Boston: Houghton Mifflin, 1979.

[13] L. G. Ungerleider and M. Mishkin, “Two cortical visual systems,” in *Analysis of Visual Behavior*, D. J. Ingle, M. A. Goodale, and R. J. W. Mansfield, Eds. Cambridge, MA: MIT Press, 1982, pp. 549-586.

[14] F. Locatello et al., “Object-Centric Learning with Slot Attention,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 11525-11538, 2020.

[15] Z. Liu et al., “Swin Transformer: Hierarchical Vision Transformer using Shifted Windows,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 10012-10022, 2021.

Appendix

Pseudocodes and equations to demonstrate the feasibility of the concept:

Spatial Consistency Loss: To enforce proprioception, the allocentric coordinates of the static background ($X^{(t)}_{bg_{a}llo}$) must remain constant despite the robot’s movement over time (t): $L_{spatial} = ¿ \vee X^{(t)}_{bg_{a}llo} - X^{(t-1)}_{bg_{a}llo} \vee ¿^{2}$

PyTorch Pseudocode (Grid-Cell Binder): The “Place Cell” block that projects objects onto the global spatial grid based on odometry using Cross-Attention.

```
import torch
import torch.nn as nn

class GridCellBinder(nn.Module):
    def __init__(self, num_grid_nodes=1024, d_model=512):
        super().__init__()
        # Learnable grid nodes of the global spatial grid (Queries)
        self.grid_nodes = nn.Parameter(torch.randn(1, num_grid_nodes, 
d_model))
        self.cross_attention = 
nn.MultiheadAttention(embed_dim=d_model, num_heads=8)

    def forward(self, entity_tokens, x_ego, T_odometry):
        # 1. Homogeneous transformation (X_allo = T_odo_inv * X_ego)
        T_inv = torch.linalg.inv(T_odometry)
        x_allo = torch.bmm(x_ego, T_inv.transpose(1, 2))

        # 2. Keys (Where is it?) and Values (What is it?)
        keys = self.allocentric_coord_proj(x_allo)
        values = entity_tokens
```

```python
# 3. Topological Binding onto the spatial grid
B = entity_tokens.size(0) # Batch size
allocentric_world_state, _ = self.cross_attention(
    query=self.grid_nodes.repeat(B, 1, 1).transpose(0, 1),
    key=keys.transpose(0, 1),
    value=values.transpose(0, 1)
)
return allocentric_world_state.transpose(0, 1)
```